Quiz :- 20

C20: $a_1 = 2$; $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$

(a)

$n=1 \Rightarrow a_1 = 2$

$n=2 \Rightarrow a_2 = \frac{1}{2}\left(2 + \frac{2}{2}\right) = \frac{3}{2}$

$n=3 \Rightarrow a_3 = \frac{1}{2}\left(\frac{3}{2} + \frac{2}{3/2}\right) = \frac{1}{2}\left(\frac{3}{2} + \frac{4}{3}\right) = \frac{17}{12}$

$n=4 \Rightarrow a_4 = \frac{1}{2}\left(\frac{17}{12} + \frac{2}{17/12}\right) = \frac{1}{2}\left(\frac{17}{12} + \frac{24}{17}\right) = \frac{577}{408}$

$n=5 \Rightarrow a_5 = \frac{1}{2}\left(\frac{577}{408} + \frac{2(408)}{577}\right) = \frac{1}{2}\left(\frac{332929 + 332928}{235416}\right)$

$= \frac{1}{2}\left(\frac{665857}{235416}\right) = \frac{665857}{470832}$

$n=6 \Rightarrow a_6 = \frac{1}{2}\left(\frac{665857}{470832} + \frac{(470832)\times 2}{665857}\right)$

$= \frac{1}{2}(1) = \frac{1}{2}$

$a_1 = 2$

$a_2 = 3/2$

$a_3 = 17/12$

$a_4 = 577/408$

$a_5 = 665857/470832$

$a_6 = 1$

$$\lim_{n\to\infty} 2 + a_{n+1} = 2 + \lim_{n\to\infty} \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$$

$$= \infty$$

∴ Diverges by geometric

(B)

$$\frac{n}{\sqrt{4n^2+8}}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{\sqrt{4n^2+8}}$$

(A)

$$= \frac{1}{(-2)(4n^2+8)^{3/2}} \cdot 8n$$

(C)(A)

$$\sum_{n=1}^{\infty} \left(\frac{5}{3}\right)^n$$

$$|r| = \frac{5}{3} > 1$$

Diverges by geometric

(B) $\sum_{n=2}^{\infty} (2/7)^n$

$|r| = 2/7 < 1$

Converge by Geometric Series.